

Cryptography and security:

With the looming threat of quantum computers breaking widely used encryption algorithms, the urgency to develop quantum-resistant encryption methods has intensified. For instance, the National Institute of Standards and Technology (NIST) has been actively soliciting proposals for post-quantum cryptography standards.

The researchers at the Massachusetts Institute of Technology (MIT) have unveiled a new quantum-resistant encryption protocol called 'Lattice-Based Encryption', which has shown promising results in resisting attacks from both classical and quantum computers.

Optimisation problems: Quantum computing excels in solving optimisation problems due to its ability to explore vast solution spaces simultaneously. Volkswagen, for instance, has been exploring the use of quantum computing to optimise traffic flow in urban areas, aiming to reduce congestion and emissions.

In 2023, D-Wave Systems announced the release of its Advantage quantum computer, boasting over 5000 qubits and improved coherence times, making it better suited for tackling real-world optimisation challenges.

Drug discovery and materials science: Quantum simulators have shown remarkable promise in accurately modelling molecular structures and properties, significantly speeding up the drug discovery process. In 2021, the collaboration between Google Quantum AI and Harvard University demonstrated the use of quantum algorithms to simulate the electronic structure of small molecules with unprecedented accuracy, paving the way for more efficient drug design. Quantum computing is also being explored in materials science for designing novel materials with tailored properties, such as superconductors and lightweight alloys.

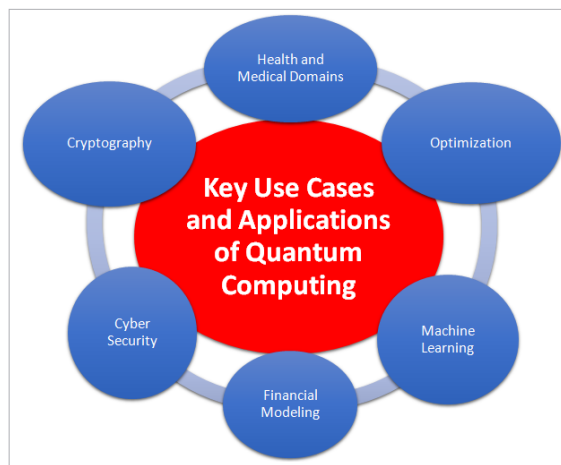


Figure 1: Key use cases of quantum computing

Milestones and developments

In August 2019, Google made headlines by claiming to have achieved quantum supremacy with its 53-qubit Sycamore processor, completing a calculation in 200 seconds that would take traditional supercomputers thousands of years.

In June 2023, researchers at the University of Science and Technology of China published a paper in the journal *Science*, detailing their achievement of quantum supremacy using a photonic quantum computer. Their device, employing 76 photons, outperformed classical computers in a specific computational task.

IBM continues to make strides in quantum computing accessibility through its IBM Quantum Experience platform, which provides cloud-based access to quantum processors and a suite of tools for quantum algorithm development.

Challenges and future outlook

Despite these advancements, challenges such as qubit stability, error correction, and scalability remain significant hurdles in the path towards practical quantum computing.

Companies and research institutions worldwide are actively working on addressing these challenges, with investments pouring into qubit technologies, error mitigation techniques,

and quantum algorithm development.

Quantum computing is expected to have a transformative impact across various sectors in the coming years, revolutionising fields ranging from finance and logistics to healthcare and materials science.

These developments underscore the growing momentum in the field of quantum computing and its potential to reshape our technological

landscape in profound ways.

Programming platforms for quantum computing based applications

Different programming platforms for quantum computing-based applications cater to different needs and expertise levels. These platforms offer tools, libraries, and resources for developing, simulating, and running quantum algorithms. Here are some notable ones.

IBM Quantum Computing Platform
<https://quantum-computing.ibm.com/>

IBM Quantum Experience provides cloud-based access to quantum computers as well as simulators. Users can write quantum circuits using Qiskit, IBM's open source quantum computing framework. It offers a range of tutorials, documentation, and community support for both beginners and experienced users. It also offers high-level abstractions for quantum circuits, algorithms, and applications. The platform includes simulators for testing quantum programs and provides educational resources such as tutorials, textbooks, and community forums.

Microsoft Quantum Development Kit
<https://www.microsoft.com/en-us/quantum/development-kit>